DRUGS · DRUGS THAT HARM THE EAR AND NOSE

Ototoxic drugs

Ototoxic drugs damage the cochlea, the vestibular system, or both. The damage may be reversible (salicylates, loop diuretics, macrolides) or permanent (aminoglycosides, platinum compounds). ENT surgeons are asked to monitor patients receiving these drugs, to recognise ototoxic hearing loss and bilateral vestibular failure, and to advise on prevention and rehabilitation.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

OTOTOXIC DRUGS

DRUG OR GROUP	MAINLY AFFECTS	REVERSIBLE?	KEY POINTS
Aminoglycosides (gentamicin, streptomycin: vestibular; amikacin, neomycin, kanamycin, tobramycin: cochlear)	Hair cells, basal turn first	Usually permanent	Mitochondrial m.1555A>G mutation; loop diuretics add risk; topical drops through a perforation
Cisplatin (and carboplatin at high doses)	Outer hair cells and stria vascularis	Permanent	Dose-related, worse in children; sodium thiosulfate protects in some children
Loop diuretics (furosemide, bumetanide)	Stria vascularis	Usually reversible	Rapid intravenous doses, renal failure; strongly synergistic with aminoglycosides
Salicylates (aspirin at high doses)	Outer hair cells (prestin)	Reversible	Tinnitus and flat mild loss at high levels

DRUG OR GROUP	MAINLY AFFECTS	REVERSIBLE?	KEY POINTS
Quinine, chloroquine, hydroxychloroquine	Cochlea	Often reversible (quinine)	Tinnitus, vertigo; cinchonism
Macrolides (erythromycin, azithromycin)	Cochlea	Usually reversible	High intravenous doses, renal or hepatic impairment
Vancomycin	Cochlea	Variable	Mainly with aminoglycosides
Deferoxamine	Cochlea	Variable	Chelation in transfusion-dependent patients
Phosphodiesterase-5 inhibitors (sildenafil)	Cochlea	Variable	Associated with sudden sensorineural hearing loss
Topical agents in the middle ear	Inner ear through the round window	Variable	Aminoglycoside drops, chlorhexidine and alcohol preparations, some antiseptics

CHAPTER TWO

2 Recognising and monitoring

What ENT surgeons do.

2.1 Monitoring

Record baseline pure-tone audiometry (with extended high frequencies 9 to 16 kHz where available, and otoacoustic emissions in children) before treatment, repeat during treatment and afterwards. Significant change by ASHA criteria is a threshold shift of 20 dB or more at one frequency, 10 dB or more at two adjacent frequencies, or loss of response at three consecutive frequencies where there was one before. Ask about tinnitus and imbalance.

2.2 Vestibular toxicity

Bilateral vestibular loss causes oscillopsia and unsteadiness, especially in the dark or on uneven ground, without vertigo. Test with the head impulse test (bilateral catch-up saccades), a positive Romberg on foam, dynamic visual acuity, and video head impulse or caloric testing. Treat with vestibular rehabilitation.

2.3 Prevention

Avoid ototoxic drugs where possible, use the lowest effective dose and shortest course, monitor drug levels and renal function, avoid combining ototoxic drugs, and test for the m.1555A>G mutation where available.

3 CHAPTER THREE For the examination

What the examiner wants to hear.

3.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Which drugs are **cochleotoxic, vestibulotoxic, reversible or permanent**.
- The **mechanisms** (hair cell entry, reactive oxygen species, stria vascularis).
- **Monitoring** criteria and how bilateral vestibular loss presents.
- The m.1555A>G mutation.

3.2 Common pitfalls

COMMON PITFALLS

- No baseline audiogram before cisplatin or long aminoglycoside courses.
- Missing bilateral vestibular failure after an ICU stay.

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